

LAB REPORT

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Database Forensics

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For the class of Computer Forensic, taught by

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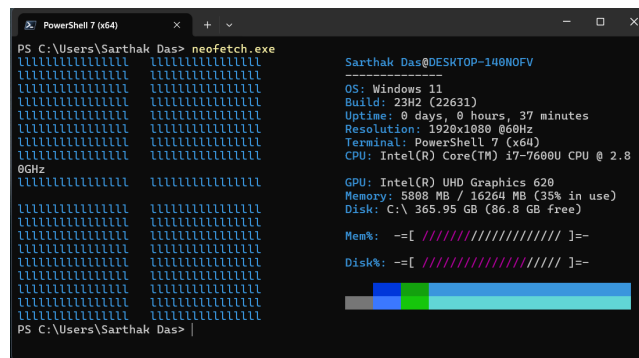
1 Introduction

1.1 Objectives

This report is a documentation of my lab work on database forensics. The primary objective of this lab is to analyze SQLite database files using DB Browser for SQLite.

1.2 System Information

For the purposes of this lab, a Lenovo ThinkPad X270 was used, having the following system specifications: Intel Core i7-7600U CPU @ 2.80GHz with Intel UHD Graphics 620, 16GB RAM, 1TB SSD running operating system Windows 11 23H2 (Build 22631).



```
PS C:\Users\Sarthak Das> neofetch.exe

Sarthak_Das@DESKTOP-140N0FV
OS: Windows 11
Build: 23H2 (22631)
Uptime: 0 days, 0 hours, 37 minutes
Resolution: 1920x1080 @60Hz
Terminal: PowerShell 7 (x64)
CPU: Intel(R) Core(TM) i7-7600U CPU @ 2.8
GHz
GPU: Intel(R) UHD Graphics 620
Memory: 5808 MB / 16264 MB (35% in use)
Disk: C:\ 365.95 GB (86.8 GB free)

Mem%:  --[  ////////////////////////////////// ]--
Disk%:  --[  ////////////////////////////////// ]--
```

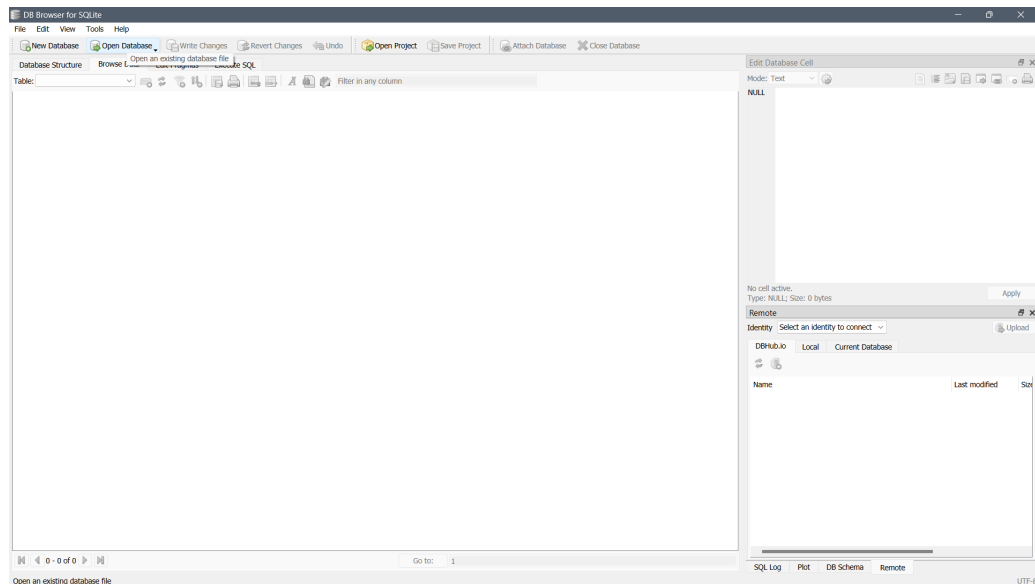
1.3 Software Used

The software used for this practical is DB Browser for SQLite Version 3.13.1.

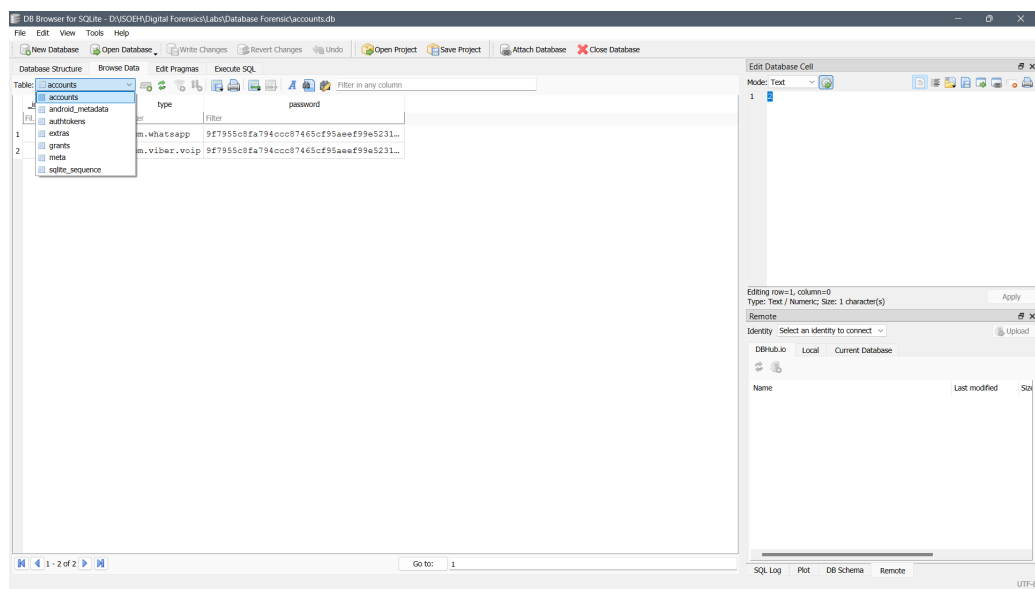


2 Objective: Database Browsing

We begin by opening DB Browser for SQLite, and selecting **Open Database**, choosing the database file we wish to browse. For the purposes of this lab, we were provided with three database files – *accounts.db*, *contacts2.db* and *browser2.db*.



Once opened, we can browse the databases to look for suspicious activity.



3 Objective: Analysis

From the *accounts.db* database, we retrieve the password hashes of the suspect's WhatsApp and Viber accounts. These can be fed into hash cracking tools to retrieve the passwords for further investigation.

From the *contacts2.db* database, we retrieve the contacts and call records of the suspect. These people can be traced down and interrogated for further investigation.

From the *browser2.db* database, we retrieve the browsing history and bookmarks of the suspect. There are several red flags, such as:

- Web searches for “murder” and “hacking tools”.
- Topics such as “hacker tools” and “make a bottle bomb” bookmarked.

All these when taken together serve as strong evidences in court of law.